

Speech Tech Week 1

AQ1.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.

1) Speech has richer information than text. So it is possible to detect _____

1 point

- ☒ speaker's emotion
- ☒ speaker's gender
- ☒ speaker's identity
- ☐ speaker's bank balance

2) Statement A: Text is another form of communication like speech.
Statement B: Speech is the easiest form of communication compared to text.

1 point

Choose the correct options from the above statements

- ☒ Both A and B are True
- ☐ Both A and B are False
- ☐ A is True but B is False
- ☐ A is False but B is True

3) Listener ear's _____ membrane excites the nerves and the speaker's message is understood by the listener's brain

basilar

1 point

AQ1.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.

1) The first step before the continuous time speech signal is fed to computers is _____

sampling

→ process is called sampling

1 point

2) For perfect reconstruction of a bandlimited signal, the sampling frequency must be _____

1 point

- ☐ equal to twice the maximum frequency
- ☒ greater than equal to twice the maximum frequency
- ☐ lesser than equal to twice the maximum frequency
- ☐ none of the above

$SF \geq 2 \times \text{Max freq.}$

3) Human ear is capable of hearing up to _____ kHz.

20

1 point

AQ1.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.

1) State True/False: Hearing involves frequency analysis.

1 point

- ☒ True
☐ False

2) According to Fourier, any periodic signal can be represented as a linear combination of complex sinusoids of different frequencies under certain conditions.

1 point

3) Fourier series expansion is useful in finding the frequency components in a periodic signal.

1 point

4) Z-Transform is a tool to analyze the interaction between systems and signals.
(Enter your answer in Comma Separated i.e., xxx,yyy)
Signals, Systems

1 point

5) Two types of Digital filters are FIR and IIR.

1 point

- ☒ FIR → finite impulse response
☒ IIR → infinite impulse response
☐ AIR
☐ IAR

AQ1.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.

1) Different speech sounds are produced by different mouth configurations.

1 point

- ☒ True
☐ False

2) One glottis time period of men is greater than the women.

1 point

- ☒ men
☐ children
 $T_{men} > T_{women} > T_{children}$

3) Voiced sounds result in periodic speech signals.

1 point

- ☒ periodic
☐ aperiodic

4) Phonemes are the alphabets of spoken language.

phonemes

1 point

AQ1.6: Activity Questions 6 - Not Graded

This assignment will not be graded and is only for practice.

1) In the terminal analogous model for speech, the impulse train input corresponds to ____ sounds.

- ☒ voiced
- ☐ unvoiced

→ voiced

1 point

2) Different resonant frequencies of the vocal tract are named as ____

- ☒ formants
- ☐ pitch

→ formants
captured by
wideband spectrum

1 point

3) State True/False: Terminal Analogous Model is also called as causal filter model.

- ☐ True
- ☒ False

→ speech productⁿ
model

1 point

4) In Short Time Fourier Transform (STFT) for speech applications, the usual window length is ____ ms.

25

overlap → 10

1 point

Practice Assignment 1 - Not Graded

1) Message formulation for speech happens in ____

brain

1 point

2) Statement A: Text is another form of communication like speech.

Statement B: Speech is the easiest form of communication compared to text.

Choose the correct options from the above statements

- ☒ Both A and B are True
- ☐ Both A and B are False
- ☐ A is True but B is False
- ☐ A is False but B is True

1 point

3) Sampling theorem was given by ____

- ☐ Nyquist
- ☐ Shannon
- ☒ Nyquist Shannon
- ☐ None of the above

Nyquist - Shannon

1 point

4) Telephone channel acts as a band pass filter

1 point

- ☐ Low pass
- ☐ High pass
- ☒ Band pass
- ☐ Band reject

5) The transform that can be performed by computers is Discrete Fourier Transform

1 point

- ☒ Discrete Fourier Transform
- ☐ Continuous Time Fourier Transform
- ☐ Any Fourier Transform

6) Determine whether the following discrete time system is linear $y[n] = e^{x[n]}$

1 point

- ☐ linear
- ☒ non-linear

7) lungs forces air through the glottis for speech production.

1 point

8) Short time Fourier Transform is required to analyze speech because speech is quasi-stationary

1 point

- ☒ quasi-stationary
- ☐ stationary
- ☐ non-stationary

9) /p/ is a stop

1 point

- ☐ glide
- ☐ nasal
- ☐ fricative
- ☒ stop

1) **State True/False.** Computers can handle only numbers. So computers cannot handle speech signal without some signal processing.

1 point

- ☒ True
- ☐ False

Yes, the answer is correct.

Score: 1

Accepted Answers:

True

2) **State True/False:** Aliasing distorts the reconstructed signal

1 point

- ☒ True
- ☐ False

Yes, the answer is correct.

Score: 1

Accepted Answers:

True

3) Determine whether the following discrete time system is memoryless $y[n] = x[n^2]$

1 point

- ☐ memoryless
- ☒ not memoryless

Yes, the answer is correct.

Score: 1

Accepted Answers:

it should depend only on the current $x[n]$, which is so not the case in $y[n] = x[n^2]$

4) Determine whether the following discrete time system is stable $y[n] = x[n^2]$

1 point

☒ stable

☐ unstable

No, the answer is incorrect.

Score: 0

Accepted Answers:

stable

if you check this is in the bounds of each other, therefore, stable

5) Wideband spectrogram has ____ time resolution.

1 point

☒ good

☐ poor

Yes, the answer is correct.

Score: 1

Accepted Answers:

good

→ good

6) State True/False: Terminal Analogous Model is also called as causal filter model.

1 point

☐ True

☒ False

No, the answer is incorrect.

Score: 0

Accepted Answers:

7) State True/False: MP3 is a lossy storage technique

1 point

☒ True

☐ False

Yes, the answer is correct.

Score: 1

Accepted Answers:

True

and they operate at lower bit rates

8) Any point in a 3-D space can be represented as a ____ combination of its basis functions

linear

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: String) linear

linear
basis function

9) The continuous time Fourier Series expansion of a continuous time and periodic signal results in a ____ and ____ output in frequency domain

1 point

(Enter your answer in Comma Separated i.e., xxx,yyy)

discrete, aperiodic

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: String) Aperiodic,Discrete

CTFS
I → C & P
D → D & A